

Scenario

A network engineer has implemented a custom topology in Mininet and wants to run a simple ping test between two hosts to verify basic connectivity, controlled by a Floodlight controller.

Question: After starting Mininet with the Floodlight controller, what would be the expected OpenFlow interaction (message types) between the hosts, switches, and the controller for the first ping packet from h1 to h2?

Answer:

When executing the first `ping` from host `h1` to host `h2` in a Mininet topology managed reactively by the Floodlight SDN controller, no flow rules exist initially in the switch's flow table. The interaction sequence across ARP resolution and the initial ICMP Echo Request follows these OpenFlow message exchanges:

1. ARP Request Handling (Host MAC Discovery):

- **Host Packet Generation:** `h1` sends an ARP Request broadcast to resolve `h2`'s MAC address.
- **OFPT_PACKET_IN :** The ingress switch checks its flow table, finds a flow miss, encapsulates the ARP Request, and sends an `OFPT_PACKET_IN` message to the Floodlight controller.
- **OFPT_PACKET_OUT :** Floodlight processes the ARP Request and sends an `OFPT_PACKET_OUT` message instructing the switch to flood the ARP Request across all active ports (or output to the specific port where `h2` resides).

2. ARP Reply Handling:

- **Host Packet Generation:** `h2` receives the ARP Request and transmits a unicast ARP Reply destined for `h1`.
- **OFPT_PACKET_IN :** The switch encounters another flow miss for this unicast ARP traffic and forwards an `OFPT_PACKET_IN` message to Floodlight.
- **OFPT_FLOW_MOD & OFPT_PACKET_OUT :** Floodlight learns `h2`'s location/port, sends an `OFPT_FLOW_MOD` message to install a reactive flow rule for traffic destined to `h1`, and issues an `OFPT_PACKET_OUT` message to forward the ARP Reply back to `h1`.

3. ICMP Echo Request (The First Ping Packet):

- **Host Packet Generation:** With ARP resolution complete, `h1` sends the first ICMP Echo Request packet targeted at `h2`.

- **OFPT_PACKET_IN** : The switch attempts flow matching for the IP packet. Finding no existing rule for this IP flow (forward path from `h1` to `h2`), it sends an `OFPT_PACKET_IN` message to Floodlight.
- **OFPT_FLOW_MOD** : Floodlight calculates the shortest path from `h1` to `h2` and sends an `OFPT_FLOW_MOD` message to install flow rules along the switch path (matching source/destination MAC and IP addresses with an action to forward out the egress port connected towards `h2`).
- **OFPT_PACKET_OUT** : Simultaneously, Floodlight sends an `OFPT_PACKET_OUT` message containing the buffered ICMP packet to deliver it immediately out the designated egress port to `h2` .

Subsequent ping packets (and ICMP Replies) between `h1` and `h2` match the installed flow rules directly in hardware, bypassing the controller entirely.